consolidated feeds at each data center to be able to compete with other competing consolidators. Therefore, the Commission believes that the geographic latency reduction in the decentralized consolidation model can be achieved even if one existing market data aggregator enters the competing consolidator business. The benefit of the decentralized consolidation model with regard to geographic latency will not rely heavily on the assumption that a large number of consolidators would enter the market.[2234] Importantly, as discussed above,[2235] geographic latency is the biggest cause of latency differentials between current SIP data distributed by exclusive SIPs and current proprietary data feeds.[2236]

Also, the Commission understands that many current market data aggregators rely on wireless communications to receive data from various exchange data centers, using fiber connections as a backup in case of bad weather. As discussed above,[2237] wireless communications are faster than current transmission methods for SIP data. To the extent that the business practices of current market data aggregators serve as a model for competing consolidators, the Commission expects the decentralized consolidation model to reduce consolidation and transmission latency as well.[2238] Additionally, some competing consolidators could achieve lower consolidation and transmission latency by processing subsets of consolidated market data for market participants that prefer narrower data content than the

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2234 See supra Section V.C.2(a); V.C.2(a)(ii).

2235 See supra Section V.B.2(b).

2236 Several commenters agreed. See, e.g., Cboe Letter at 23; Nasdaq Letter IV at 49; STANY Letter II at 6.

2237 One commenter agreed and provided a technical explanation for these speed differentials. See, e.g., Data Boiler Letter I at 39. See also supra Section V.B.2(b) for a discussion on the current process for dissemination of SIP data and proprietary data feeds.

2238 Some commenters agreed. See, e.g., ICI Letter at 10.

entirety of consolidated market data. The Commission believes that the effect of the decentralized consolidation model on the consolidation and transmission latencies depends on robust competition among competing consolidators going forward.

The Commission believes that to the extent that the benefits of faster access to market data come from the ability to engage in more timely participation in the provision of liquidity, this effect represents an economic benefit to the equity market generally because it will provide more fair and equal access to market data and will reduce information asymmetries among market participants.[2239] In particular, to the extent that the existing advantages of having access to fast proprietary data feeds are derived from trading strategies exploiting differentials in the speed of access to market data (i.e., exploiting traders in the market who currently rely solely on slower SIP data), this benefit would represent a transfer from current users of faster proprietary data to the users of consolidated market data products in the decentralized consolidation model that will now also have access to faster data.[2240]

In order for both economic benefits and transfers to be realized, at least some market participants that are new users of fast and more content-rich consolidated market data products will need to possess the technological capability to take advantage of the speed improvements

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2239 One commenter agreed and said that "[l]ow latency proprietary traders with independent decision engines in different data centers will always be the fastest actors in the system; however, lessening the information asymmetry between these actors and other market participants has great value." See Capital Group Letter at 4.

2240 One commenter agreed and stated that "[r]ace conditions are impossible to solve. Even if you're fastest by a picosecond, you are still first." See Nasdaq Letter III at 5. For a discussion of the effect of speed differentials on trading, see also Don Bollerman, *A NYSE Speed Bump You Weren't Aware Of*, IEX available at https://www.sec.gov/comments/10-222/10222-395.pdf (last accessed Jan. 8, 2020).

the decentralized consolidation model is likely to provide. It is the Commission's understanding that such technological capabilities are costly to acquire, and this fact could reduce the amount of benefit and the degree to which individual market participants can profit (through the transfers mentioned above) from the decrease in data latency.

Several commenters disagreed with the Proposing Release's predictions on latency reduction.[2241] Some commenters stated that the existence of multiple competing consolidators will not reduce latency much because processing times are already minuscule.[2242] Other commenters argued that additional latency gains are unlikely to improve outcomes for retail and long-term investors.[2243] Other commenters argued that competing consolidators will be an extra hop on the data delivery chain and market participants receiving data from competing consolidators will always be slower than self-aggregators or proprietary data users who receive market data directly from the exchanges.[2244]

The Commission believes that the decentralized consolidation model will reduce latency rates for market data and will bring consolidated market data products' latency rates more in line with the latency rates of proprietary data feeds. This latency reduction could come from all

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2241 See, e.g., Nasdaq Letter IV at 36; Cboe Letter at 23; Citadel Letter at 5; Nasdaq Letter III at 5; STANY Letter II at 6; Data Boiler Letter II at 1; NBIM Letter at 6.

2242 See, e.g., Nasdaq Latter IV at 49; Cboe Letter at 23; Citadel Letter at 5; STANY Letter II at 6.

2243 See, e.g., Nasdaq Letter IV at 41; Proof Trading Letter at 1.

2244 See, e.g., Nasdaq Letter IV at 8; Healthy Markets Letter I at 3–4; Data Boiler Letter II at 1.

forms of latency, including geographic latency.[2245] Even if the potential gains from processing speeds are small, competing consolidators could achieve larger latency reductions by decreasing geographic latency. Unlike the exclusive SIPs, competitive forces will incentivize competing consolidators to respond to market participants' needs. For example, for market participants whose trading strategies depend on low-latency data, some competing consolidators could create an instance of consolidated market data in every data center, significantly reducing geographic latency.[2246] Furthermore, while retail and long term investors might have less latency sensitive trading strategies, even small gains in speed can be meaningful for improving execution quality, a benefit to investors.[2247] Finally, the Commission expects the introduction of the decentralized consolidation model to reduce data latency for market participants who currently rely on SIP data but will switch to using consolidated market data products, because competing consolidators will be incentivized to provide faster data products than the exclusive SIPs. The Commission discusses the full details of the relationship between self-aggregators and competing consolidators with respect to latency below.[2248] This then will lead to a reduction in information

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2245 Several commenters agreed. See, e.g., DOJ Letter at 3–4; IntelligentCross Letter at 4; Wellington Management Letter at 1; BlackRock Letter at 5.

2246 Several commenters agreed. See, e.g., BlackRock Letter at 5; Wellington Letter at 1. Additionally, one commenter stated that "[g]eographic latency could be addressed either through a distributed SIP or competing consolidators, therefore agreeing with the Commission. See Nasdaq Letter IV at 49.

2247 See supra Section V.B.3(e) for discussion of latency and execution quality.

2248 See infra Section V.C.4(b) for this discussion.

asymmetry caused by current large latency differences among investors using SIP data versus proprietary data feeds.[2249]

One other commenter disagreed with the Commission's assessment of the decentralized consolidation model's latency benefits, stating that "[c]competing consolidators will create a costly arms race in speed."[2250] The Commission believes that there is already demand for fast data in the market and the introduction of the decentralized consolidation model will not affect market participants' demand for faster data. However, the new model will affect the supply of market data speeds available to market participants. Hence, with the amendments, market participants who currently rely on SIP data will have other data options that are faster than SIP data and that are potentially a closer substitute to proprietary data feeds.

(iv) Market Resiliency

Fourth, the Commission believes that the decentralized consolidation model will eliminate the single point of failure in market data consolidation and dissemination step and potentially increase market resiliency. However, the Commission acknowledges that the provision of data content underlying consolidated market data will continue to be a single point of failure, in that one of the exchanges could experience a systems issue leading to a market-wide effect just like they could today if they experience a systems issue when delivering their data content to the exclusive SIPs.

Under the amendments, with the availability of multiple competing consolidators, there could be multiple copies of consolidated market data, which will contribute to market

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2249 Some commenters agreed. See, e.g., Capital Group Letter at 4; AHSAT Letter at 3.

2250 See Angel Letter at 19.

resiliency.[2251] Some commenters stated that having multiple competing consolidators will reduce the probability of market-wide failures and instead increase market resiliency.[2252] The Commission agrees. Currently, each exclusive SIP consolidates and disseminates unique market data and if either or both of the exclusive SIPs experience a systems problem the whole market is affected. However, under the amendments, with multiple competing consolidators serving the market, no single competing consolidator's systems issue will be a market-wide problem. At most, it will affect all of its customers or some of its customers if others retained a back-up competing consolidator.

Other commenters stated that competing consolidators will move the market from single point of failure to multiple points of failure and reduce resiliency.[2253] The Commission agrees that with multiple competing consolidators, each of their systems issues might cause problems for a certain portion of the market participants. However, that will still decrease the probability of market-wide failures in data consolidation and dissemination because all competing consolidators would have to have a simultaneous systems issue for there to be a market-wide failure in data consolidation and dissemination. That is unlike today, when a single exclusive SIP's systems issue can create a market-wide failure in data consolidation and dissemination. Additionally, with multiple competing consolidators, market participants will have a choice if

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2251 See infra Section V.C.2(e)(i) for a discussion of how Regulation SCI could also contribute to market resiliency.

2252 See, e.g., BlackRock Letter at 5; Committee on Capital Markets Regulation Letter at 3; Clearpool Letter at 7–8; BestEx Research Letter at 5. See also Section III.C.2.

2253 See, e.g., NYSE Letter II at 24; Nasdaq Letter IV at 7, 8, 36; Cboe Letter at 23–24, 25; TechNet Letter II at 2; Data Boiler Letter II at 4.

they decide to retain a back-up competing consolidator based on their business needs. However, currently, for market participants that primarily rely on SIP data there is no secondary back-up consolidator option. Thus, as discussed above,[2254] the Commission does not believe that the decentralized consolidation model reduces resiliency.

Finally, one commenter expressed concerns about low-cost providers being less resilient.[2255] The Commission believes that low cost data providers would not necessarily be less resilient and, if any are less resilient, that would not necessarily lead to lower resiliency in the market because market participants could review competing consolidators' monthly disclosures and decide whether to retain a backup consolidator. First, any low cost competing consolidators that are above the five percent (5%) market data revenue threshold will be subject to Regulation SCI with geographically diverse backup requirements. Second, all competing consolidators are required to publicly disclose, on a monthly basis, their system availability and performance statistics. In a competitive market, this will encourage competing consolidators to invest in their systems to make sure that they have high rates of system "up-time." Additionally, it will give market participants information to anticipate their backup needs and decide whether they need to get a backup consolidator based on their data providers' system availability and performance statistics.

(d) Costs of the Decentralized Consolidation Model

The Commission believes that the amendments are likely to have direct costs on potential competing consolidators and SROs, and indirect costs on existing exclusive SIPs, certain market participants and investors, and SROs. As explained below, the Commission estimates that the

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2254 See supra Section III.C.2.

2255 See Nasdaq Letter IV at 35.

direct costs to each potential competing consolidator will be between approximately $5.6 million in ongoing annual costs, and total one-time costs of up to between approximately $1.7 million and $5.7 million, depending on entity type.[2256] Further, the Commission estimates that SROs will jointly have approximately $175,000 in direct one-time costs and approximately $102,000 in ongoing costs for the amendments to the effective national market system plan(s). Each SRO will also incur approximately $71,000 in one-time direct costs and approximately $128,000 in ongoing costs for the collection and dissemination of information. The Commission expects, however, that the amendments that introduce a decentralized consolidation model will have additional indirect costs. Some of these direct and indirect costs are likely to be passed on to investors in terms of the prices they will pay.

(i) Direct Costs to Potential Competing Consolidators

As mentioned in the Proposing Release and discussed above,[2257] the Commission believes that five types of entities may register to become competing consolidators and will have to build systems, or modify existing systems, that comply with the rules: (1) market data aggregation firms, (2) broker-dealers that currently aggregate market data for internal uses, (3)

2256 These costs do not include the costs of compliance with Regulation SCI, which are discussed below. See infra Section V.C.2(e)(ii). The direct cost of compliance with Regulation SCI (i.e., PRA plus non-PRA costs) is approximately between $1 million and $2.4 million in ongoing costs and is approximately between $300,000 and $3 million in one-time costs, depending on entity type. Therefore, the total direct cost of the decentralized consolidation model, including the costs of compliance with Regulation SCI is approximately between $6.6 million and $8 million in ongoing costs and is approximately between $2 million and $8.4 million in one-time costs, depending on entity type. However, these costs could be lower for some competing consolidators that choose not to take in and offer the entirety of consolidated market data as well as for some that do not have to comply with Regulation SCI.

2257 See Proposing Release, 85 FR at Section V.D.2; supra Section IV.D.3.

the existing exclusive SIPs, (4) new entrants, and (5) SROs. The Commission estimates that all direct ongoing annual costs and some one-time costs will be common among all competing consolidators and that some one-time costs will vary depending on entity type.

For purposes of the PRA,[2258] the Commission estimates that direct ongoing costs for each competing consolidator will be approximately $5.63 million and consist of the following costs: costs of $16,812 to amend Form CC prior to the implementation of material changes to pricing, connectivity, or products as well as to correct inaccurate or incomplete information;[2259] costs of $50 to obtain digital IDs for the purposes of signing the Form CC annually,[2260] costs of approximately $5.56 million associated with operating and maintaining a competing consolidator system;[2261] costs of $362 to ensure that it has posted the correct direct URL hyperlink to the Commission's website on its own website;[2262] costs of $4,360 of recordkeeping;[2263] and costs of $45,222 to prepare and make publicly available a monthly report.[2264]

2258 Direct costs cited in this section are quantified from estimates in the PRA. See supra Section IV.

2259 See supra Section IV.D.1(b)(iii); supra note 1412.

2260 See supra Section IV.D.1(a).

2261 These costs are composed of labor costs of $418,290, external costs of $371,175 to operate and maintain systems to comply with Rules 614(d)(1) through (d)(4), external costs of $168,000 to purchase market data from the SROs, and an additional annual ongoing external cost of $4,602,720 to co-locate itself at four exchange data centers. See supra Section IV.D.3(g)(iii).

2262 See supra Section IV.D.2(b)(iii); supra note 1423.

2263 See supra Section IV.D.4(b)(iii).

2264 See supra Section IV.D.5(b)(iii); supra note 1543.

The Commission estimates that direct one-time costs that are common across all competing consolidators will be $189,342 and consist of the following costs: costs $93,540 to complete an initial Form CC;[2265] costs of $50 to obtain digital IDs the purposes of signing the initial Form CC;[2266] costs of $5,604 to file material amendments to Form CC;[2267] costs of $121 to publicly post the Commission's direct URL hyperlink to its website upon filing of the initial Form CC;[2268] costs of $8,720 to keep and preserve at least one copy of all documents made or received by it in the course of its business and in the conduct of its business;[2269] costs of $80,507 to produce the monthly reports and costs of $800 for an external website developer to create the website that will post and keep the monthly reports.[2270]

The Commission estimates that the total direct costs to each market data aggregation firm or each broker-dealer that currently aggregate market data for internal uses that will decide to register as a competing consolidator will include approximately $5.63 million in ongoing annual costs, as discussed above, and total one-time costs of approximately $1.71 million. The one-time costs are composed of labor costs of $697,150;[2271] external costs of $618,750 to modify its systems to comply with Rules 614(d)(1) through (d)(4), external costs of $14,000 to purchase market data from the SROs, an additional initial external cost of $194,000 to co-locate itself at

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2265 See supra Section IV.D.1(a); supra note 1402.

2266 See supra Section IV.D.1(a).

2267 See id.

2268 See supra Section IV.D.2(a)(iii).

2269 See supra Section IV.D.4(a)(iii).

2270 See supra Section IV.D.5(a)(iii).

2271 See supra Sections IV.D.3(b)(iii), IV.D.3(c)(iii).